

# Xiaoqing PAN

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Leibniz Institute on Aging - Fritz Lipmann Institute

Re: Postdoc Microbiome in Stroke and Aging

Dear Dr. Winek,

My name is Xiaoqing, I just finished my PhD study at Hans-Knöll-Institute in Jena, Germany. I expect to have my defense in July 2026. My PhD was on double-stranded RNA metabolism in both humans and fungi. What draws me to your lab is that your group is now investigating microbiota-gut-brain communication via next-generation sequencing methods.

I studied microbiology in the context of human fungal pathogens. I was studying how *Aspergillus fumigatus* produces small RNAs, how dicer-like proteins process dsRNA substrates, and how the resulting 21-nt 5'-U-containing sRNAs are loaded into argonaute complexes. I analyzed sRNA-seq and mRNA-seq data to characterize the small RNA landscape and the contribution of RNAi machinery to fungal biology. After the bioinformatic prediction, I further use qPCR as well as Northern blot for the experimental validation. Work also involved in tRNA fragment functional analysis and sRNA trafficking in between fungi and humans via extracellular vesicles. Through this project, I studied how to grow microorganisms, RNA-seq analysis and wet-lab experiments. Work has resulted in two publications on *RNA* and *Molecular microbiology*. In parallel, I also participated in a project investigating transcriptomic responses in human PBMCs. Through characterizing transcriptomic responses after fungal stimulations with a focus on host alternative splicing and A-to-I RNA editing, we surprisingly found out that one fungus can induce RNA editing as well as disrupt host RNA splicing. Work has gained me experiences analyzing multiple RNA-seq cohorts, DEG analysis via DESeq2, alternative splicing analysis via rMATS-Turbo, RNA structure analysis via SHAPE-Map and RNA editing analysis.

I am interested in human gut microbiota because one of the fungal pathogens I studied is commensal and can naturally live in a human body. On the computational side, I built pipelines on HPC for RNA-seq, sRNA-seq, and tRNA-seq, and I have experience analyzing sequencing data from both host and microbial fractions within the same experiment. On the wet-lab side, I have hands-on experience with sRNA cloning, custom sequencing library preparation, gene expression analysis by qPCR and Northern blot, cultivation of human cell lines. Additionally, I also have teaching experience in microbiology and molecular biology together with my supervisor Dr. Blango and other colleagues.

The bigger picture of this project is to investigate brain-body communication in stroke using multiple sequencing data sets. I can learn biomedicine, flow cytometry techniques. I am aware that your current sequencing data is focusing on bacteria, but checking on fungal species might also be a good option, which I might be helpful. At the same time, I am seeking an international collaborative work environment, as I find integration into a vibrant team important for my development and performing high-quality science. FLI is a big institute in Germany. I also value the interdisciplinary nature of your team. I noticed your group includes members with diverse backgrounds, and I believe that bringing my perspective from fungal genetics and host-pathogen immunology could contribute fresh insights to your projects. In the future, I want to improve my writing skills for a grant (e.g., Walter Benjamin Programme).

Thank you for your time and consideration.

Best,

Xiaoqing

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## SUMMARY

RNA biologist and computational transcriptomics researcher with expertise in high-throughput sequencing, gene regulation, and RNA-mediated cellular responses. Experienced in integrating molecular biology with bioinformatics to study small RNAs, RNA interference, alternative splicing, and RNA editing across host and microbial systems. Proficient in RNA-seq analysis, reproducible computational workflows (R, Bash), and sequencing-based approaches for investigating post-transcriptional regulation.

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## TECHNICAL SKILLS

- **Molecular Biology:** sRNA cloning, NGS library preparation, GUS reporter assays, luciferase and GFP reporter assays, Gibson Assembly Cloning, Genetic Engineering, Northern Blot, qPCR, PCR, immunostaining, LC-MS, cultivation of microorganisms (BSL-1/2), plant cultivation and tissue preparation
- **Bioinformatics:** RNA-seq, sRNA-seq, differential expression, alternative splicing, small RNA characterization and target prediction, data visualization, HPC (Slurm), UCSC, SwissProt, KEGG, NCBI
- **Programming:** R, Python, Bash, Git, Docker, Linux
- **Language:** Chinese (native), English (fluent), German (B1)

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## EXPERIENCES

### RNA landscape of human fungal pathogen *Aspergillus fumigatus*

- Characterized RNAi and tRNA fragments processing pathways in *A. fumigatus* across developmental stages (conidia, mycelium), revealing morphotype-specific RNA regulatory mechanisms
- Analyzed differential expressed genes on mRNA-, sRNA-seq datasets, uncovering post-transcriptional regulatory networks controlling fungal development and stress response.
- Validated genes of interest (sRNA, tRNA fragments, sRNA targets) via qPCR, northern blot

### Human PBMC transcriptome changes upon fungal stimulations

- Analyzed pathogen-specific transcriptomic signatures in human immune cells responding to fungal pathogens, characterized differential expressed genes, alternative splicing, A-to-I RNA editing, pathogen recognition receptor pathway activation
- Built HPC pipeline, wrote custom R scripts for multi-condition cohort analysis

### Human neutrophils exporting extracellular RNA upon *A. fumigatus* confrontation

- Analyzed EV-associated sRNA, tRNA, and mRNA cargo from primary human neutrophils during fungal challenge

### Non-coding RNA analysis of rice-*Xanthomonas oryzae* interaction

- Studied how T3SS-dependent bacterial effectors from *Xanthomonas oryzae* shape host non-coding RNA regulation; performed GUS reporter assays, plant cultivation, tissue preparation, LC-MS, qPCR, and PCR
- Engineered overexpressed and knockout mutants of circular RNA and microRNA in rice, established genetic tools that enabled investigation of ncRNA-mediated regulation

## Teaching

- Microbiology and Molecular Biology, MMB005, 2022-2025. Supervised master students investigating gene regulation together with Dr. Blango and other colleagues.

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## EDUCATION

**PhD** | Microbiology, Friedrich Schiller University, Germany (supervisor: Dr. Matthew Blango) 2026

*Thesis: RNA regulation in host-fungal interactions: investigation of the host epitranscriptome and fungal small RNAs*

**MSc** | Bioengineering, Chinese Academy of Sciences, China (supervisor: Dr. Kuaifei Xia) 2021

*Thesis: Non-coding RNAs ciR133 and miR168a-5p play roles in response to Xoo and abiotic stress in rice, respectively*

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## PUBLICATIONS

- **Pan X\***, Kelani AA\*, Schrettenbrunner L, Prabakar S, Israni B, Blango MG. (2026). An improved description of the small RNA landscape of the human fungal pathogen *Aspergillus fumigatus*. Mol Microbiol 1-12.
- **Pan X\***, Bruch A\*, Blango MG. (2024). The past, present, and future of RNA modifications in infectious disease research. ACS Infect Dis 10(12), 4017-4029.
- Xia K\*, **Pan X\***, Chen H, Xu X, Zhang M. (2023). XOO-responsive transcriptome reveals the role of the circular RNA133 in disease resistance by regulating expression of OsARAB in rice. Phytopathol Res 5, 22.
- Xia K\*, **Pan X\***, Zeng X, Zhang M. (2023). Rice miR168a-5p regulates seed length, nitrogen allocation and salt tolerance by targeting OsOFP3, OsNPF2.4, and OsAGO1a, respectively. J. Plant Physiol 208, 153905.
- Kelani AA, Bruch A, Riviaccio F, Visser C, Krüger T, Weaver D, **Pan X**, Schaeuble S, Panagiotou G, Kniemeyer O, Bromley MJ, Bowyer P, Barber AE, Brakhage AA, Blango MG. (2023). Disruption of the *A. fumigatus* RNA interference machinery alters the conidial transcriptome. RNA 29(7), 1033-1050.

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## CONFERENCES

- **2025 - Oral presentation:** German speaking Mycological Society (DMyKG), Germany
- **2025 - Poster:** Gordon Research Seminar - RNA Editing, Italy
- **2024 - Oral presentation:** Jena RNA Club Symposium, Germany
- **2024 - Poster:** RNA Society Meeting, UK
- **2023 - Poster:** EMBL - The Non-coding Genome Symposium, Germany

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## AWARDS & SCHOLARSHIP

- DAAD Scholarship 2021 - 2025
- Jena School for Microbial Communication (JSMC) travel grant 2023

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## REFEREE

- Dr. Matthew Blango ([matthew.blango@leibniz-hki.de](mailto:matthew.blango@leibniz-hki.de))
- Dr. Olaf Kniemeyer ([olaf.kniemeyer@leibniz-hki.de](mailto:olaf.kniemeyer@leibniz-hki.de))